VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

VK26-IITI-AHM-0295

Project Overview

Title - ‘SwasthEdge’

Team ID - VK26-IITI-AHM-0295

Track - Theme: AI in Hardware Manufacturing

Sub-Theme: Edge AI & On-Chip Processing

“A low-cost self-diagnosing edge controller that fuses sensor, electrical, and controller-health data to confirm faults locally and trigger appropriate responses without continuous cloud connectivity.”

Problem Statement

Operational Pain Point -

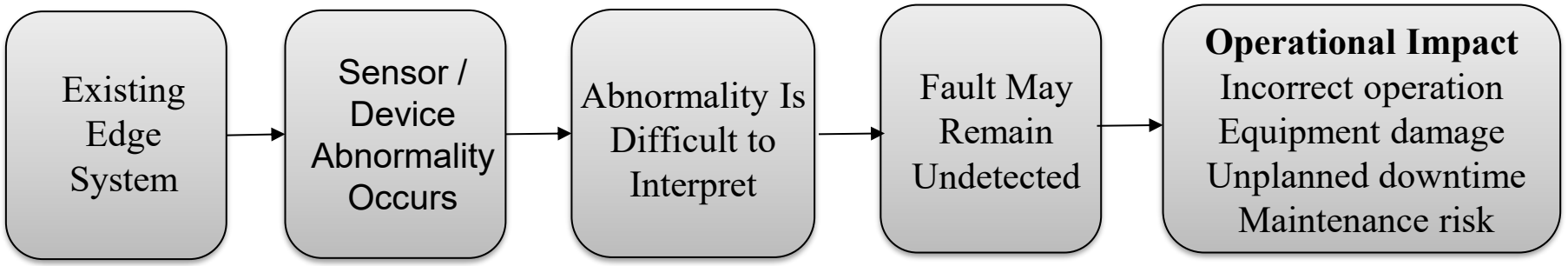
Low-cost edge controllers in industrial and remote environments can face sensor drift, disconnection, abnormal power consumption, and supply instability, creating a need for an embedded system that can monitor device and sensor health, distinguish persistent faults from temporary abnormalities, and provide an appropriate local response.

Target Beneficiary Segment -

Small and medium-scale industries, manufacturing systems, smart agriculture, environmental monitoring, remote infrastructure, and Industrial IoT/edge deployments where continuous network connectivity or manual monitoring may not always be available.

User Journey – Before State

Flowchart



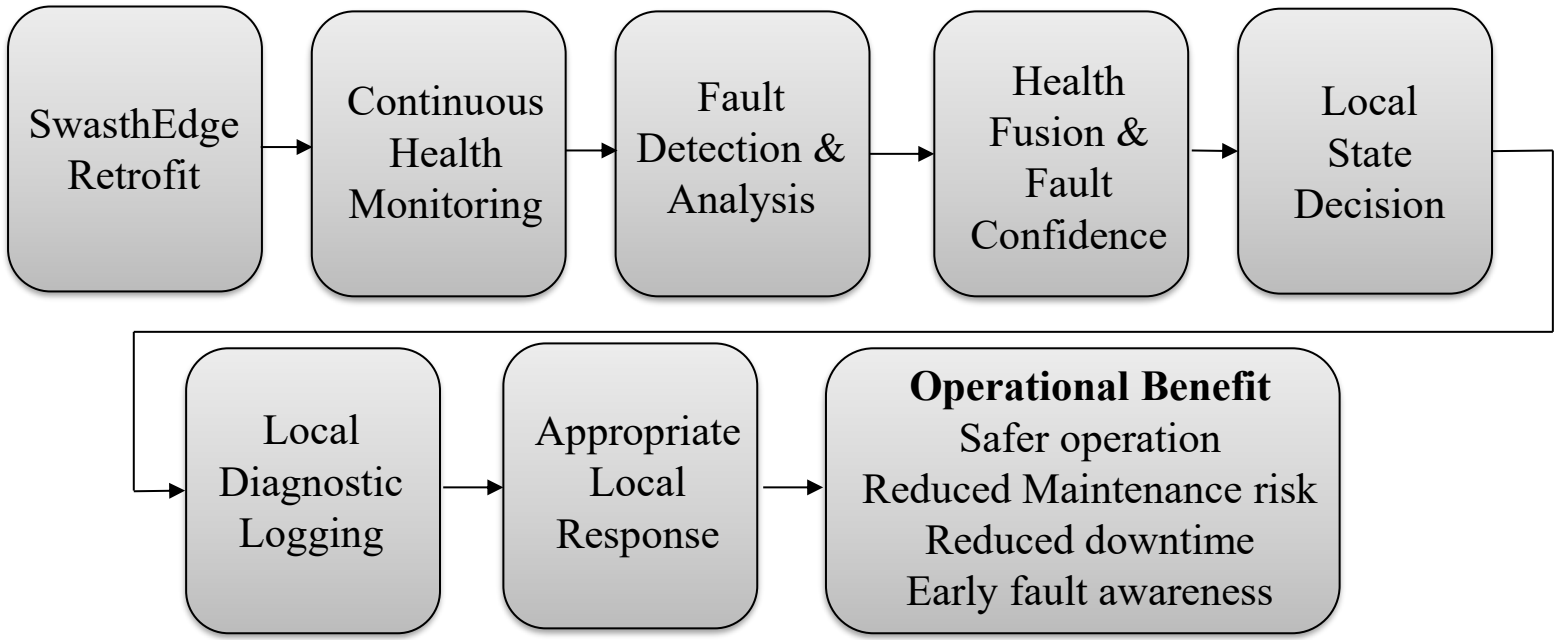
BASELINE INEFFICIENCY — TO BE ESTABLISHED

Fault-Detection Latency	False-Alarm Behaviour	Processing Overhead	Operational Impact
Time taken to recognize an injected fault	Incorrect responses to temporary abnormalities	CPU / RAM / processing cost	Effect of undetected or delayed faults

Baseline values will be established through controlled fault-injection tests before SwasthEdge evaluation.

User Journey – After State

Flowchart

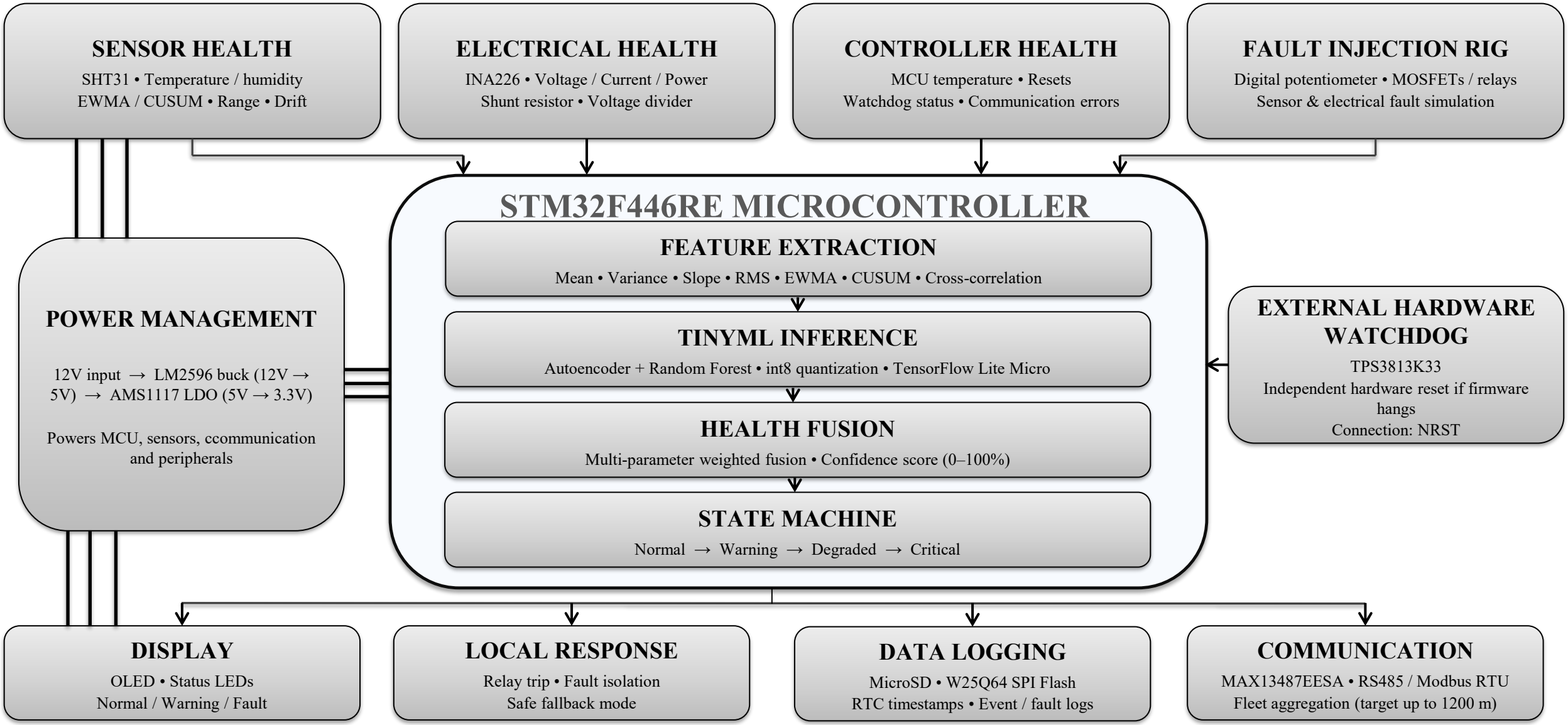


EFFICIENCY METRICS — TO BE BENCHMARKED

Fault-Detection Latency	Fault-Confirmation	Processing & Power Overhead	Local Response & Continuity
Time from fault injection to local detection	Ability to distinguish persistent faults from temporary abnormalities	CPU / RAM / processing latency / additional power	Response state and diagnostic retention during network loss

SwasthEdge performance will be compared with sensor-only detection using controlled fault-injection tests.
From reactive detection to locally confirmed, self-diagnosing operation

System Block Diagram



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Target Specifications & Hardware Selection

Parameter	Target Specification	Rationale
Inference Latency	< 100 ms	Ensures real-time detection before critical damage.
Flash Budget	< 80 KB (of 512 KB)	Proves viability on a cost-sensitive, resource-aware MCU.
RAM Budget	< 32 KB (of 128 KB)	Leaves headroom for other application tasks.
Current Draw	< 20 mW incremental	Maintains energy efficiency for remote edge deployments.
Fault States	4-State Model	Enables graceful degradation (Normal → Critical).
Comms Range	Up to 1200m	Ensures fleet aggregation in large factories without Wi-Fi.

Subsystem	Selected Hardware	Interface / Protocol
Core Controller	STM32F446RE (Arm Cortex-M4, 180 MHz)	—
Sensor Health	SHT31-DIS (Temperature & Humidity)	I2C (400 kHz)
Electrical Health	INA226 (Voltage, Current, Power)	I2C (400 kHz)
Fleet Comms	MAX13487EESA (Isolated RS485)	Modbus RTU (UART)
Local Logging	W25Q64 SPI Flash & MicroSD Card	SPI
Safety / Watchdog	TPS3813K33 (External Hardware Watchdog)	GPIO / NRST
Actuation	5V SPDT Relay & ULN2003 Driver	GPIO
Power Management	LM2596 (Buck) & AMS1117 (LDO)	12V → 5V → 3.3V

❖ **Architecture Rationale:-**

STM32F446RE DSP support enables on-device TinyML without an external NPU. I²C reduces sensor pin count; isolated RS485 supports long-distance communication in electrically noisy environments. The TPS3813 watchdog remains independent of firmware to recover from lockups.

PCB

3D Design View

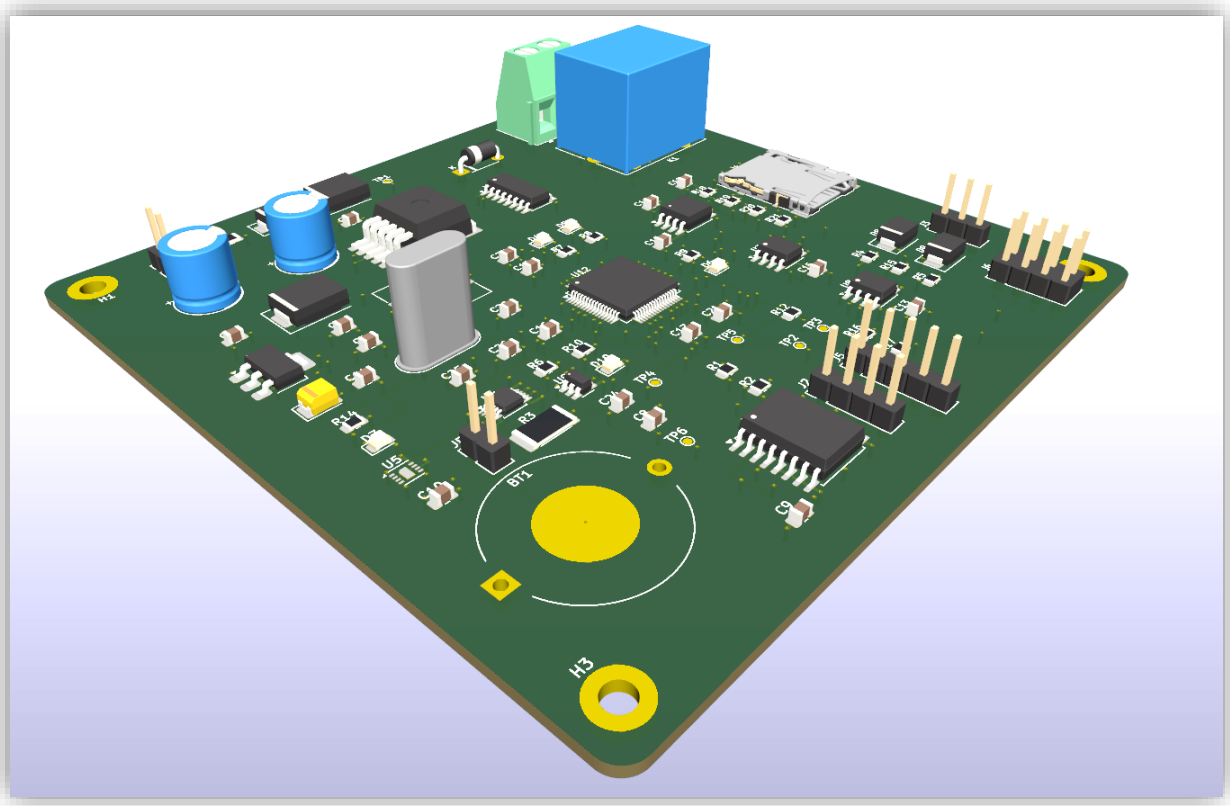


Figure: Custom 120x100mm 2-layer PCB integrating sensing, communication, and fault-injection circuitry for cost-sensitive viability.

Physical Specifications

120 × 100 mm board footprint	2-layer FR-4 PCB	1.6 mm board thickness

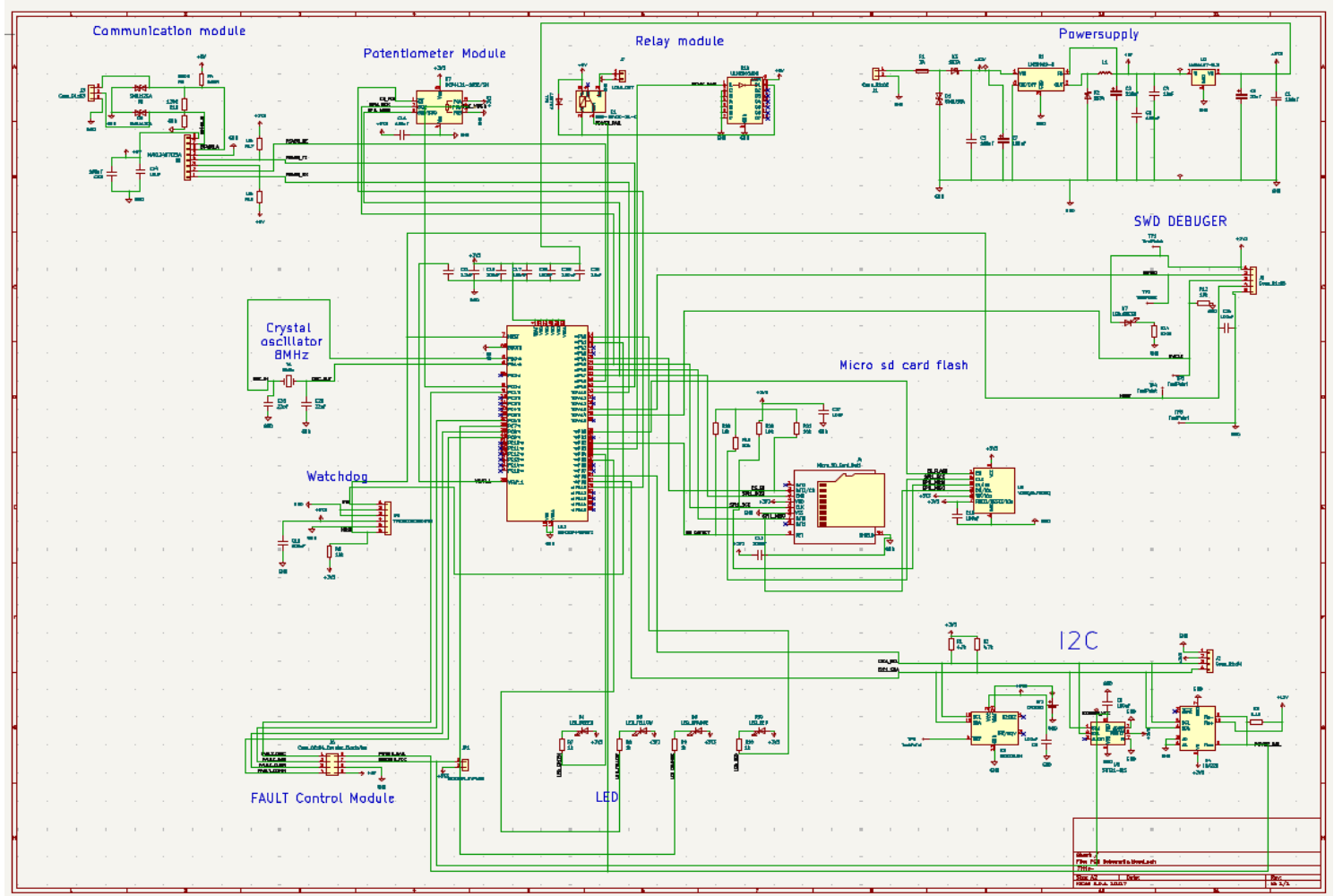
FUNCTIONAL ZONES REPRESENTED IN THE LAYOUT

Functional Zone	Description
CORE COMPUTE	STM32F446RE MCU and 8 MHz crystal clock.
SENSING & COMMS	I²C sensor path plus isolated RS485 fleet interface.
RESILIENCE	External watchdog, protection devices and fault-injection circuitry.
DATA & POWER	W25Q64 SPI flash, microSD interface, LM2596 buck and AMS1117 LDO.

DESIGN INTENT	A compact, low-cost board integrates monitoring, communications and deliberate fault stimulus on one controller platform.

Electronics Schematic & PCB Integration

KiCAD Schematic



SCHEMATIC SUBSYSTEM MAP

Subsystem	Description
MCU + CLOCK	STM32F446RE with 8 MHz crystal and debug header.
SENSORS + I²C	SHT31-DIS and INA226 share the internal sensor bus.
FLEET COMMUNICATION	MAX13487EESA isolated RS485 interface for Modbus RTU.
LOCAL DATA	W25Q64 SPI flash and microSD card interface.
SAFETY + FAULTS	TPS3813 watchdog, reset path, protection and fault-injection header.
POWER RAILS	12 V input conversion to 5 V and 3.3 V rails.

❖ **Design Rationale:-**

The schematic integrates independent health-monitoring mechanisms (I2C sensors, RS485, Watchdog) around a single low-cost STM32F446RE MCU. The dedicated fault-injection module allows us to physically induce and test faults on the running device, proving the system's autonomous response.

Bill of Materials

Component	Part / Specification	Quantity	Lead Time	Estimated Cost (INR)
Core Controller	STM32F446RET6 (Arm Cortex-M4, 180MHz)	1	Local (3 days)	₹550
Health Sensors	SHT31-DIS (Temp/Humidity) & INA226 (Power)	1 Set	Domestic (7 days)	₹700
Comms & RTC	MAX13487EESA (RS485) & DS3231SN (RTC)	1 Set	Domestic (5 days)	₹380
Data Logging	W25Q64 (8MB SPI Flash) & MicroSD Slot	1 Set	Local (2 days)	₹130
Power & Relay	LM2596, AMS1117, 5V Relay, ULN2003	1 Set	Local (2 days)	₹250
Fault Injection	TPS3813 Watchdog, MOSFETs, TVS Diodes	1 Set	Domestic (5 days)	₹350
Passives	8MHz Crystal, Capacitors, Resistors, LEDs	1 Set	Local (2 days)	₹250
Custom PCB	FR4 2-Layer 1.6mm (120x100 mm)	5	Domestic (6 days)	₹1,500

PROCUREMENT STATUS

Core parts are listed as locally or domestically sourced within India; the maximum stated shipping lead time is 7 business days.

COST-SENSITIVE VIABILITY

Under the **₹4,200** target in the supplied project narrative, keeping electronics costs accessible for MSME retrofits.

Maker Journey



Figure: Team reviewing electronic design and development work together

Stage 1 to Stage 2 Engineering Progress & Evidence

- **SYSTEM & FIRMWARE**

  STM32F446RE-based edge controller; firmware assets organized in the repository.

- **ELECTRONICS DESIGN**

  KiCad schematic, PCB layout and bill of materials are grouped under /hardware.

- **TINYML WORKFLOW**

  ML data, notebooks and model assets are organized under /ml.

- **DOCUMENTATION & DEMO**

  Architecture, validation documents, presentation material and demo-video links are grouped in the repository.

Work-In-Progress Software Evidence

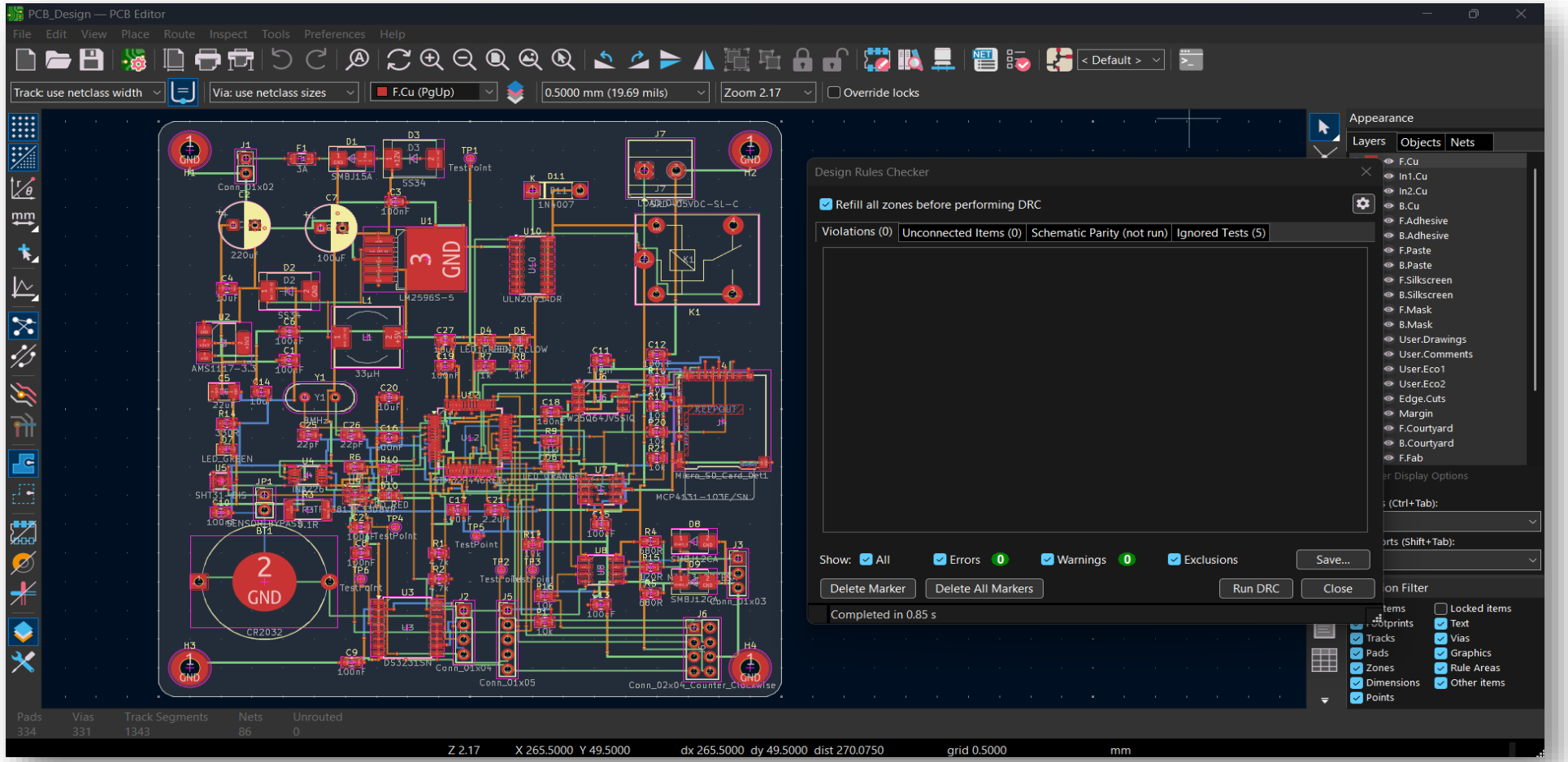


Figure - EDA layout software (KiCad)

RAW REPOSITORY LINK	github.com/ConquerToFreedom/SwasthEdge
	Firmware • Hardware • ML • Documentation • Presentation • Demo resources

Safety & Hazard Mitigation

Fault Confirmation
Use sensor readings together with voltage, current and MCU-health indicators before increasing fault confidence.

Graceful Degradation
Normal → Warning → Degraded → Critical operating states prevent unnecessary escalation.

Local Fail-Critical Response
Warning, faulty-sensor isolation, fallback values, reduced operation, communication recovery, or predefined critical state.

Fault Monitoring
Monitor sensor drift, power abnormalities, controller health and communication failures continuously.

Standards / Compliance

Functional Safety — IEC 61508
Reference for safety-related electrical/electronic/programmable functions and their safety lifecycle.

Industrial EMC — IEC 61000-6-2 / IEC 61000-6-4
Reference for industrial-environment immunity and emissions of electrical/electronic equipment.

Industrial Cybersecurity — IEC 62443-4-1 / IEC 62443-4-2
Reference for secure product development and technical security requirements for industrial automation/control-system components.

Standards are considered as design and validation references; no formal compliance or certification is claimed at the prototype stage.

Roadmap

Fabrication Phase (Months 1 - 2)

October 2026 — System Setup & Breadboard Prototype

- Finalize MCU, sensor and power-monitoring architecture
- Establish normal sensor/controller behaviour
- Build breadboard-based functional prototype
- Integrate voltage/current and sensor monitoring
- Prepare controlled fault-injection setup

November 2026 — Self-Diagnosis Firmware

- Implement sensor-drift detection using EWMA/CUSUM
- Implement MCU health and communication monitoring
- Integrate local diagnostic logging
- Evaluate lightweight TinyML-based anomaly detection
- Prepare firmware for PCB migration

Integration & Field Demo (Months 3 - 4)

December 2026 — PCB & Health-Fusion Integration

- Finalize schematic and PCB layout
- Develop health-fusion and fault-confidence logic
- Implement Normal → Warning → Degraded → Critical states
- Integrate local fault responses and fallback handling
- Assemble and bench-test PCB prototype

January 2027 — Validation & Stage 3 Readiness

- Conduct controlled fault-injection tests
- Compare sensor-only vs. multi-parameter health fusion
- Measure detection performance, false alarms, latency, CPU/RAM and power overhead
- Validate PCB prototype and Stage 3 field-testing readiness

Stage 3 Target: Working SwasthEdge prototype validated through controlled fault injection and ready for functional field testing.

Technical Challenges & Risks

Engineering Challenge / Risk	Potential Impact
Distinguishing Drift from Normal Variation	Temporary fluctuations may be misclassified as persistent sensor faults, causing false alarms.
Multi-Parameter Health Fusion	Combining sensor, power, and MCU-health indicators while maintaining reliable fault confidence can increase algorithm complexity.
Limited MCU Resources	EWMA/CUSUM, health-fusion logic, logging and continuous monitoring must remain within available CPU, RAM, flash and power budgets.
Abnormal Power / Supply Conditions	Voltage or current disturbances may affect both the monitored sensor and controller, making fault attribution difficult.
Communication Loss	Network failure can interrupt remote monitoring and requires reliable local diagnostic logging and recovery logic.
Graceful Degradation Decisions	Selecting the correct transition between Normal → Warning → Degraded → Critical without unnecessary shutdown is a key control challenge.

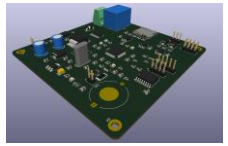
Key Engineering Constraint: Achieve reliable local self-diagnosis while keeping computation, memory, latency and additional power consumption suitable for a low-cost MCU.

Potential Solutions & Risk Mitigation

Potential Risk / Failure	Potential Solution / Mitigation
Sensor Drift	Detect gradual deviation using EWMA/CUSUM and validate against other health indicators.
Sensor Disconnection / Failure	Identify abnormal readings or loss of data, isolate the faulty sensor, and use a fallback value where applicable.
Power Instability / Abnormal Current	Continuously monitor voltage and current and raise fault confidence when electrical behaviour becomes abnormal.
MCU / Controller Fault	Monitor temperature, reset cause, watchdog, communication and memory health; move to a predefined safe state when required.
Communication Failure / Network Loss	Attempt communication recovery and use local diagnostic logging so fault information is retained without network connectivity.
Persistent vs. Temporary Abnormality	Use health fusion + fault-confidence rather than relying on a single threshold.
Escalating Fault / Unsafe Operation	Apply graceful degradation: Normal → Warning → Degraded → Critical.

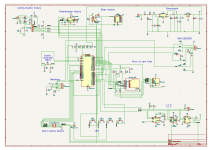
Fallback strategy: When the primary signal or subsystem becomes unreliable, SwasthEdge progressively reduces risk through validation, isolation, fallback operation, recovery attempts, and safe-state transition.

References & Sources



https://drive.google.com/file/d/18aVEA84JQQvSDFFv31FtURB8URP6s71-/view?usp=sharing

Source: kicad.org



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Image Sources



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